

SIMULTANEOUS *Go* VIA QUANTUM COLLAPSE

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ABSTRACT. We construct a simultaneous combinatorial game *SGo* that is in a precise mathematical sense a simple symmetric simultaneization of the classical board game *Go*, where simplicity is also a certain mathematical property. We develop some general theory of this symmetric simultaneization here, in particular on the universal level, and this may be of independent interest. In contrast, we prove a theorem that chess has no simple simultaneization at all. *SGo* is in some ways a simpler game than *Go*, as Komi, Ko and suicide rules are removed. On the other hand it has similar dynamics, enabled by a certain “quantum state” reduction, so that state evolution is deterministic. Using the argument of Nash, we show that a finite symmetric simultaneous combinatorial game has a mixed equilibrium strategy that draws on average, and so *SGo* does as well.

1. INTRODUCTION

Go is an ancient game governed by local-to-global geometric principles. It is a sequential game (that is, turns alternate between the two players), which has the benefit of a very simple theoretical structure, (mostly) fitting into the subject of combinatorial game theory. However, this structure also has drawbacks: we need an ad hoc rule such as Komi to deal with the advantage of Black having the first turn. Moreover, the Ko situation and its cousins like triple Ko necessitate certain non repetition rules. Although non repetition rules are necessary in the end game of many types of games, like chess for example, in Go they are necessary even in the early game, which is to say the least inelegant. Furthermore, this leads to Ko fights, in which the game loses its local-to-global character¹.

At the same time, partly due to its local-to-global nature *Go* is almost simultaneous. We make this precise by proving that *Go* has a simple (in a certain mathematical sense) symmetric simultaneization, see Theorem 26. The construction of a particular such simultaneization, called *SGo*, is the main thrust of this note. In contrast, we show that this fails for chess: augmented chess has no simple simultaneization at all, see Theorem 30. *SGo* is a combinatorial symmetric simultaneous game, which is playable on a standard *Go* board, although we need a third type of stones that we call red. It is in some ways a simpler game than *Go* as it loses the Komi, Ko and suicide rules, and it has just two rules: stone placement and capture (aside from the end of game rule). Its game dynamics are very similar to the classical *Go*.

The idea for *SGo* is loosely inspired by quantum mechanics. The rough idea is to branch the game state whenever the moves of the two players do not commute or otherwise interfere in classical *Go*. This is part of a more general theory of universal

¹In the sense that if a player engages in a Ko fight, they must make moves that are strategically disconnected.

simultaneization of certain sequential games developed in Section 4.3, and this may be of independent interest. However, the universal simultaneization generally has uninteresting dynamics, cf. Section 2.2. To get interesting dynamics in the case of *SGo*, we also have “objective reduction of the quantum state” – it is a deterministic, partial collapse of branches based on objective board information. The term objective reduction is coined by Penrose, see for instance [13]. Accordingly, a position of *SGo* carries its set of branches with it, as a position of chess carries its castling rights, see Section 2. Of course, the word quantum is used figuratively: there is no unitary time evolution, no complex amplitudes, and no probabilistic outcomes. The game is deterministic and it would not be appropriate to call it “quantum Go”. For games with genuine quantum features see for instance [6], and for a quantum version of *Go* itself [15].

Using the famous idea of Nash in [11], we show that *SGo* has a mixed equilibrium strategy to draw on average. That said, finding this even approximately is likely only practical on very small boards, due to the unreal size of the strategy space. We can also ask if *SGo* has a strategy to draw, not just to draw on average; this is true for instance on a 2×2 board.

The classical game *Go* has inspired a wealth of research on combinatorial game theory, see for instance [2], [17]. It also led to Conway’s theory of surreal numbers [4]. There is also research on simultaneous and synchronized combinatorial games, see for instance [3], [9]. We expect that *SGo* will provide a simple but rich example that can help with further study of such games.

2. GAME RULES

The game *SGo* is initialized with an empty standard *Go* board, with horizontal/vertical lines whose intersections we just call intersections. There are two players that we call Black and White. As in *Go*, players Black and White move on a given turn by placing a black respectively white stone on an empty intersection on the board, but in *SGo* they do this on the same turn without knowledge of the other’s move on this turn. We allow pass moves as in *Go*. Simultaneous action requires modifying the placement and capture rules of *Go*, while removing the Komi rule, Ko, and suicide rules. We also need to introduce a third type of stones, called red, which may be thought of as simultaneously white and black.

2.1. The entanglement state. As with chess, whose position includes the castling and en passant rights in addition to the piece diagram, a position of *SGo* carries information beyond the stone diagram: its **entanglement state**. This is a finite, nonempty set of classical *Go* positions, called **branches**, representing the classical positions that the play so far may have produced. On every turn each branch evolves as a classical *Go* position: the two moves of the turn are played on it in both orders, giving up to two new branches per branch, where a move whose intersection is occupied is forfeited (the player effectively passes in that branch). The resolution rules below then prune branches, using only visible board information as triggers.

The board that the players see and play on is the **projection** of the entanglement state: an intersection is occupied if and only if it is occupied in some branch, and a stone is displayed black respectively white if it has that color in every branch containing it, and red if it is black in some branch and white in another. So a red stone is one that is simultaneously white and black, over the branches. Captures that occur in every branch are automatically reflected in the projection and require

no rule; the role of the resolution rules below is to prune branches – to collapse entanglement – based on objective board information. In practice the entanglement state is small, and the play in Section 3 can be followed on the board alone. The formal treatment is given in Section 4.3, where Theorem 26 becomes essentially a matter of the definitions.

2.2. Motivation for the rules. Before we proceed to the main rules, we briefly overview the motivation and constraints we impose. We require:

- (1) *SGo* forms a simple simultaneization of *Go*, in the precise sense of Theorem 26. Being a simultaneization loosely means that *SGo* “embeds” into the universal simultaneization of *Go*. For example, the “causal reduction rule” in Section 2.6 is forced by the necessity of satisfying this condition.
- (2) It is playable on a standard *Go* board, which in particular requires us to represent simultaneous moves to the same position, and we choose to do this by introducing red stones. In this case there is no real choice, as any two faithful ways to represent such simultaneous action are clearly equivalent. We are representing the branched state where White moves to an intersection and Black passes, and Black moves to that intersection and White passes.
- (3) We want the minimum amount of rules, in particular we don’t want ad hoc rules dealing with Ko type situation.
- (4) The last point is at the moment less precise, but would be interesting to study more closely in the future. We want *SGo* to have similar game dynamics to the standard *Go*, that is with nearly optimal play the positions should tend to become very sharp, that is have high sensitivity. This is far from the case for the universal simultaneization $\text{Sim}(Go)$ of *Go*, as considered in Section 4.3. For $\text{Sim}(Go)$ games tend to be drawn due to players lacking forcing options.

To achieve these dynamics, we want to “optimally” prune branched histories resulting from simultaneous action. This has the effect of giving players very forcing options, see Section 3.2 for some examples. We do this via our capture/placement resolution rules, which use objective board information. But this is not without constraints, we need to prune branches so that the game stays symmetric, while having a well defined deterministic state evolution, as elaborated by Lemma 6. We also cannot prune too much, for example in the suicidal capture condition in Definition 3 we intentionally preserve certain branching to avoid the Ko type infinite loops, also see Remark 4.

2.3. Some definitions. Given a board state S , that is any (at the moment no legality restrictions) collection of white, black, red stones on the board, let $G(S)$ denote the graph with vertices stones, and an edge between vertices whenever the corresponding stones are horizontally or vertically adjacent. (Meaning they are adjacent and on the same horizontal/vertical line.) From now on adjacency always means this type of adjacency. In what follows the terms stones and vertices may be used synonymously.

A **unicolor component** C is a connected component of the full (a.k.a. induced) subgraph of $G(S)$ whose vertices are either all the black, or all the white stones. (Full subgraph means that all the edges of the original graph, between any pair

of vertices of the subgraph, are included.) In this case we call C **black**, respectively **white**. We may less often refer to **red components**, analogously defined as connected components of red stones.

As in classical *Go*, a connected component C **has no liberties** if it is not adjacent to an empty intersection (this is short for C does not contain a stone adjacent to an empty intersection). We will also call C a **trapped component**.

We define stones **adjacent** to C as those stones whose corresponding vertices are connected by an edge to C in $G(S)$.

A white/black stone on the board is called a **trapping stone** if it is adjacent to a trapped unicolor component of opposite color.

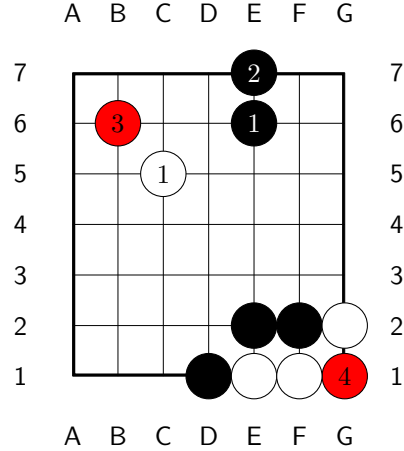
2.4. Stone placement resolution rule of *SGo*. If the players move to the same intersection i on the board, we place a stone there whose color is resolved in the following decreasing order of priority:

- (1) If either a white or black stone placed at i is trapping, then a red stone is placed at i . We call this a *trapping move*.
- (2) If the intersection i is adjacent to a black stone, and not a white stone, a black stone is placed. If the intersection is adjacent to a white stone and not a black stone, a white stone is placed.
- (3) Otherwise, a red stone is placed.

In terms of the entanglement state: each branch gains the stone of one of the two players (the two interleavings coincide here, as the second move to i is forfeited). If a red stone is placed, both kinds of branches are kept; if the stone resolves to black respectively white, only the branches in which the black respectively white move was made are kept.

Remark 1. We could do something simpler and always place a red stone in the case of simultaneous movement to i . However, as discussed in Section 2.2 we want to aggressively prune some branched histories, using objective board information.

To the right is an example of placement resolution. On move two both players move to E7, the resulting stone resolves to black by our rules. Note that the stone placed on move 4 does not resolve to white, as it is a trapping move. (One can check that no capture then occurs on this turn: the red stone blocks both parts of condition 3 for the white component $e1, f1$.)



2.5. Capture rule of *SGo*. After both White and Black move on the current turn numbered n , using the stone placement resolution rule above we get some board state S .

Definition 2. We say that a black/white/red component C is *trapped on turn n* , if C is trapped and some stone adjacent to C or in C was placed on turn n . A stone is called *trapped* on turn n , if it is an element of C as above. We say that a red stone is *trapped by white respectively black on turn n* , if replacing it by a black respectively white stone it is trapped on turn n . For a black respectively white stone, trapped by white respectively black on turn n will simply mean trapped on turn n .

Definition 3. A white component C is resolved as *captured on this turn* if the following is satisfied:

- (1) C is trapped on this turn.
- (2) If C is adjacent to a red stone that is itself adjacent to another trapped component, then that component is white. (Without this condition Lemma 6 does not work, and the game evolution will be ill defined.)
- (3) One of the following holds:
 - (a) No black or red stone adjacent to C is trapped by white on this turn.
 - (b) (Suicidal capture) No stone in C is created on this turn. If a stone in C is adjacent to a stone trapped on this turn, then that stone is created on this turn and is not red.

Remark 4. *The suicidal capture property may appear complicated, but is driven by our requirements, as outlined in Section 2.2. For example, if we remove the part of the condition “is not red”, then Ko-style infinite loops will become possible later. If we remove the condition “no stone in C is created on this turn”, then Theorem 26 will fail. For example, suppose that White and Black simultaneously place stones w and b on adjacent intersections, so that each stone is the only liberty of the other, with the remaining neighbors of w forming an old black wall, and those of b an old white wall. In classical Go the two orders of these moves disagree: the second placed stone captures the first. So this branching must be preserved in SGo, and indeed with the above condition neither component is captured and both stones remain on the board. Without it, w and b would both be captured, resulting in a state that occurs in neither branch.*

In the case C is captured:

- The stones of C are removed as prisoners of Black as in Go.
- Any red stones adjacent to C resolve as black. (Concretely, they are replaced by black stones.)

A stone converted this way is not considered to be placed or created on this turn; it retains the placement turn of the red stone it replaces.

If C is a black component the process is naturally mirrored.

In terms of the entanglement state, the resolution of C as captured is the pruning of every branch in which some stone of C survives; the removal of C and the conversion of the adjacent red stones is then just the change of the projection. For this to be meaningful we add the proviso: a component resolves as captured only if there is at least one branch in which no stone of C survives; otherwise the position remains entangled and nothing is removed.

The capture rule is applied as follows. All the unicolor components captured on this turn are resolved simultaneously: the corresponding branches are pruned, so that on the board the stones of these components are removed and the red stones adjacent to them are converted, as described above (Lemma 6). This simultaneous

resolution is then repeated, on the resulting board state, until no unicolor components captured on this turn n remain.

2.6. Causal reduction. At the end of the capture procedure, we further simplify the position as follows. If any trapped red component C is adjacent to at least one white stone but to no black stones, and none of these white stones are trapped, then all stones of C are replaced by white stones. (Requiring an adjacent white stone excludes the degenerate case of a board fully covered by red stones.) This is naturally mirrored in the case C is adjacent only to black stones. The motivation for the term ‘causal reduction’ is explained in Section 3.2.1.

Remark 5. *In terms of the entanglement state, the causal reduction prunes the branches in which stones of C are black. In a typical position no such branch exists – it would contain a black component without liberties – and then the rule is automatic: the projection already displays C white. The rule is logically necessary in the following sense: without it the board would fail to be the projection of its branches, and condition 3 of Definition 22 would fail for the map ρ in the proof of Theorem 26, as the image of ρ would contain a position that is not legally reachable.*

2.7. Stage one reduction. The above capture procedure followed by causal reduction is called **stage one reduction** and the resulting board state will be denoted by $R_n(S)$.

Lemma 6. *The stage one reduction is well defined. More precisely: the prunings of the entanglement state are unambiguous set operations, and by the capture proviso they leave the entanglement state nonempty; moreover, no red stone is adjacent to components of both colors captured on this turn, so that the displayed conversion of red stones is unambiguous.*

Proof. The first statement holds by construction. For the second, suppose that a red stone r is adjacent to unicolor components C_0, C_1 , both captured on this turn, with C_0 white and C_1 black. As C_1 is captured, it is in particular trapped. So C_0 is adjacent to the red stone r , which is itself adjacent to the trapped component C_1 of the opposite color, and Condition 2 fails for C_0 . This contradicts the assumption that C_0 is captured.

Thus, each red stone adjacent to some component captured on this turn is adjacent to captured components of only one color, and so its conversion is unambiguous. As the removal of stones is likewise unambiguous, each simultaneous resolution step, and hence the stage one reduction, is well defined. $\square$

If there are no unicolor components trapped on the turn $n - 1$ or lower, the turn is resolved. Otherwise, we recursively define higher stage reduction as follows.

Define stage k reduction by the condition that the board state after stage k reduction, denoted by $R^k(S)$, satisfies:

$$R^1(S) = R_n(S).$$

$$R^k(S) = R_{n-k+1}(R^{k-1}(S)).$$

If there are no unicolor components trapped on the turn $n - k$ or lower, the capture stage is resolved after stage k reduction.

Remark 7. *In practice the resolution of most turns is easy to work out mentally, we will give some examples of such “nested entanglement states” further ahead. Of course having a computer work out turn resolution automatically is very helpful.*

2.8. **End of the game.** An *SGo* game ends if one of the following holds:

- The players agree to end the game, in this case they may also agree to remove some black/white stones on the board as prisoners for White/Black. (As is common in standard play of *Go*.) Each black or white prisoner gives one point to the player who holds it. This is the main way of ending the game.
- The board position is unchanged after each of two consecutive turns. For example, both players perform a single stone suicide on the same turn twice in a row, or pass twice in a row. (This is meant to ensure that a winning player can actually force a win when the opponent is stalling.)
- The board has no empty intersections. (In this case only pass moves are available, so the game would in any case end two turns later by the previous condition, with the same outcome.)

We then decide the winner using Chinese *Go*-style rules (this is for simplicity). So the winner is decided by the integer sum of the prisoner count and area count. We now explain the latter.

We say that an empty intersection i on the board ***reaches black respectively white*** if there is a connected component of empty intersections (as usual meaning a connected component of the graph with vertices empty intersections on the board and edges adjacencies) that contains i and contains an intersection adjacent to a black stone respectively white stone.

We say i is ***surrounded by black*** if:

- i reaches black.
- i does not reach white. (To emphasize it could reach a red stone.)

We define surrounded by white analogously. Then in the case of player Black the area count is the sum of:

- (1) The number of black stones on the board.
- (2) The number of empty intersections surrounded by black.

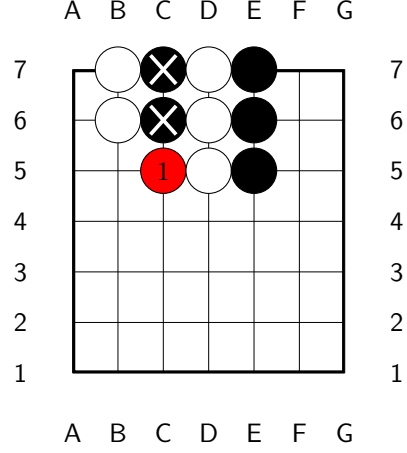
See Section 3.4 for an example.

A draw is now possible, although compared to chess it must be much less likely (on a typical 13×13 or 19×19 *Go* board).

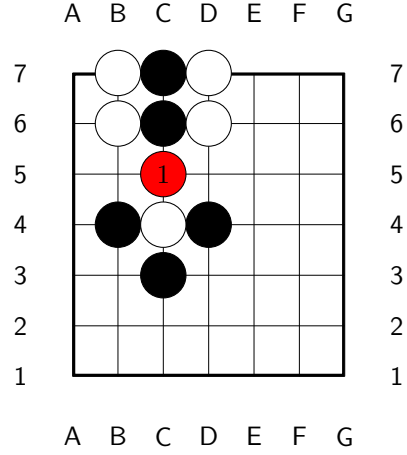
3. EXAMPLES

3.1. **Capture and non-capture moves.** Here are some examples and non-examples of capture.

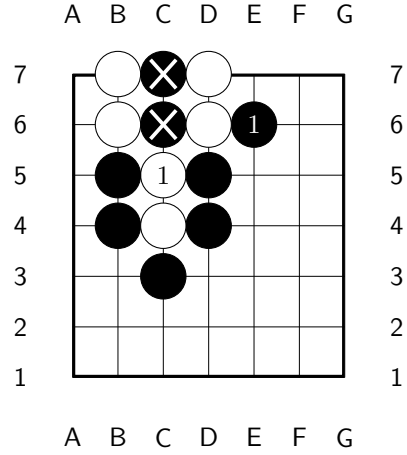
On move 1 both Black and White move to C5. As this is a capture move, C5 will resolve to white at the end of the turn.



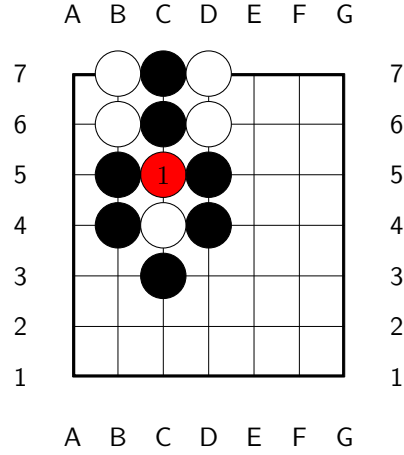
Not a capture move, as the created red stone is adjacent to an opposite-colored trapped component. This should make intuitive sense since any capture would be ambiguous from a classical *Go* perspective.



White's move is a capture move, as Condition 3b is satisfied. This is consistent with classical *Go* behavior that suicidal captures resolve to capture. Formally, this is forced by the condition that *SGo* is a simultaneization of *Go*, Theorem 26.

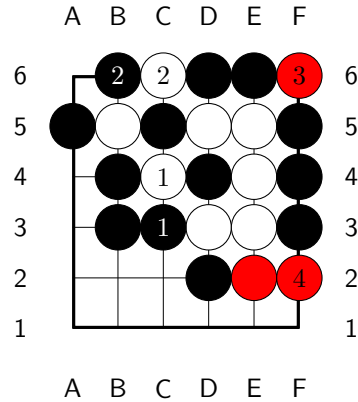


Not a capture move, now Condition 3b is not satisfied, since the stone placed on this turn is red.

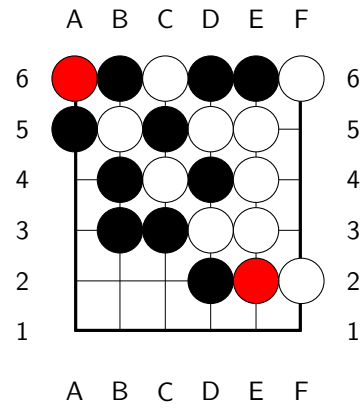


3.2. More complex examples. In the first example White needs only play with a pure strategy (and still be ostensibly winning). In sharper, more complex positions, the players may need mixed strategies.

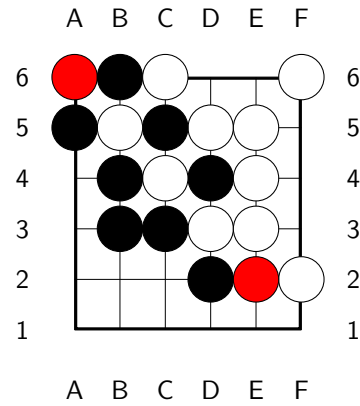
In the position on the right, on turn 1, White attempts capture at C4, while Black attempts capture at C3. Nothing is captured on turn 1 according to our rules, and after turn 2 the position becomes an example of a “nested entangled state”. A possible continuation is indicated. Note that F6 is not a capture for White under our rules, similar to the last example of the previous section. But F2 is a capture move for White since condition 3a is satisfied.



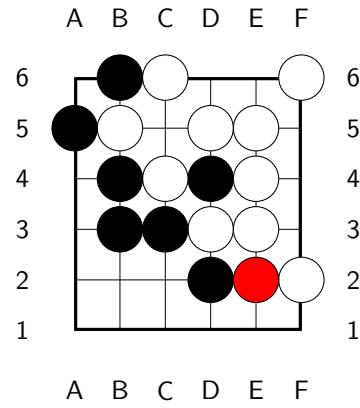
At the end of turn 4, stage one resolution is depicted on the right.



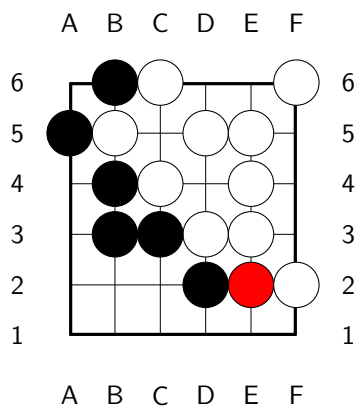
Stage two resolution.



Stage three resolution.

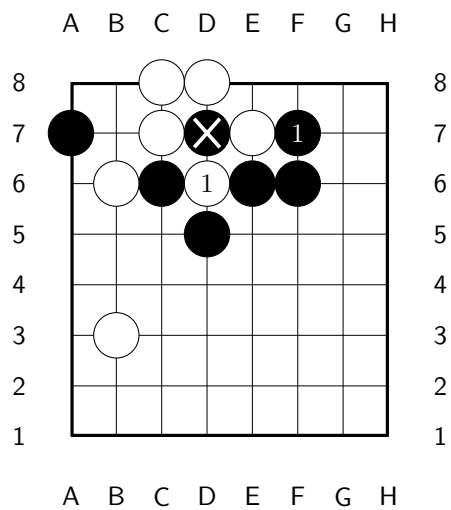


Stage four and final resolution of turn 4.

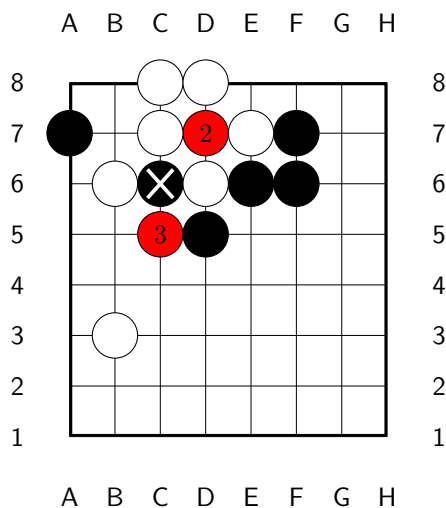


3.2.1. *Ko situation.* Below is a Ko type situation. Recall that we don't have a Ko-rule, but it is now unnecessary, since the board state cannot immediately loop by re-

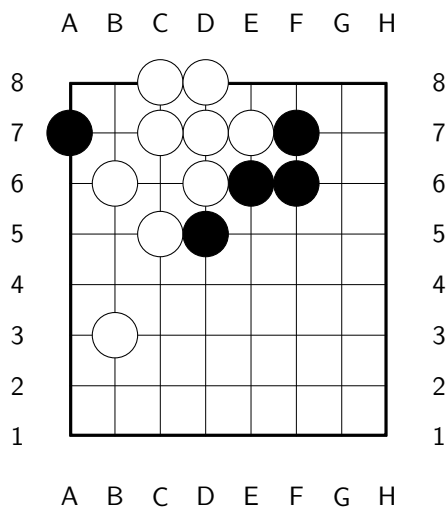
capture. White's D6 is a capture move.



Here is a possible continuation.

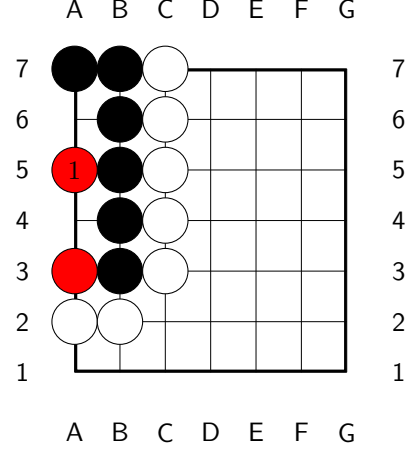


At the end of turn 3 we get the following position. The red stone at D7 is converted to white by the causal reduction rule. This should make sense from the perspective of viewing SGo as Go with quantum-like features. A red stone at D7 is meant to represent both black and white, but if it were black, the resulting position would be illegal in Go, so it must actually be white. (This is the meaning behind the term causal reduction.)

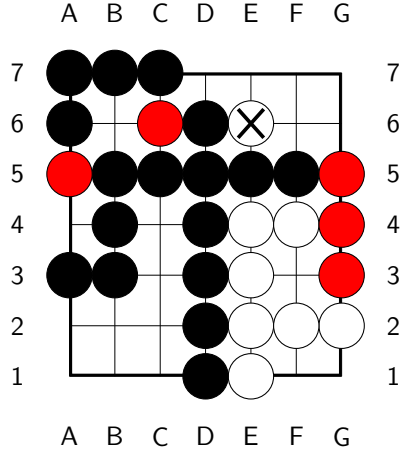


3.3. Life and death. Life and death is positionally very similar to *Go*, here is an example.

The black group is alive, that is impossible to kill.



3.4. End game position. In the diagram below, the players have agreed to end the game. The white stone at *e6* is agreed to be a prisoner of Black. Assuming there are no other prisoners, White has $7 + 3$ points and Black has $7 \cdot 7 - (7 + 3) - 5 + 1$ points, where the -5 accounts for the five red stones, which count for neither player, and the $+1$ for the prisoner.



4. SGO AS A SYMMETRIC SIMULTANEOUS COMBINATORIAL GAME

We will use the formalism of automata as it is very flexible. The goal is to introduce symmetric simultaneous combinatorial games. Such formalisms are standard in several communities: games with simultaneous moves appear as the deterministic case of the stochastic games of Shapley [19], and as the concurrent game structures of computer science [1]. Simultaneous play of combinatorial games as such is studied

by Huggan–Nowakowski–Ottaway [8], see also the references therein; there, however, the effect of a pair of simultaneous moves must be specified by each ruleset separately, no general procedure being available. The emphasis here, on simultaneousizations of a given sequential game and on their simplicity, appears to be new, and in this setting the pseudo pass moves together with the operation $\#$ further below do furnish such a general procedure. The inverse passage, representing a sequential game within an intrinsically simultaneous formalism, is standard in general game playing, and is likewise effected by pass-like moves: there the idle player is required to play an explicit “noop” move, cf. [20].

Definition 8. A *combinatorial simultaneous game* (with a pseudo pass move) G with 2 players P_0, P_1 consists of:

- A deterministic automaton G , whose set of states is a set $S(G)$ called the set of game states of G .
- The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \cup M_1(G)$: interpreted as possible moves of the players.
- $M_0(G) \cap M_1(G)$ has one move denoted as 0 and called the *pseudo pass move* (it has multi grading both 0 and 1).
- Game states decompose as

$$S(G) = B(G) \sqcup B(G) \times M(G),$$

- where $B(G)$ is some set (for example it might be the set of board states).
- Nonempty subset $F(G) = W_0(G) \sqcup W_1(G) \sqcup D(G) \subset S(G)$ of final (accepting) states, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in B(G)$, understood as the state in which the game is initialized.
- A partial game evolution function:

$$E = E_G : S(G) \times M(G) \rightarrow S(G),$$

which has the properties:

- (1) $E(s, m)$ is undefined for s in one of the final states.
- (2) If $s \in B(G) \subset S(G)$, then $E(s, m) = (s, m)$ for all $m \in M(G)$ s.t. $E(s, m)$ is defined.
- (3) If s is not final $E(s, 0)$ is defined.
- (4) If $(s, m) \in B(G) \times M(G)$, then $E((s, m), m') \in B(G)$ if defined, and is undefined if the grading of m is the grading of m' (and if neither m nor m' are 0).
- (5) Suppose that $m_0 \in M_0(G)$, $m_1 \in M_1(G)$, $s \in B(G)$, $E(s, m_0)$ is defined and $E(s, m_1)$ is defined then $E(s, m_0, m_1)$ is defined.
- (6) For s, m_0, m_1 as in the previous property if $E(s, m_0, m_1)$ is defined then so is $E(s, m_1, m_0)$ and

$$E(s, m_0, m_1) = E(s, m_1, m_0).$$

We recursively define:

$$E(s, m_1, \dots, m_k) = E(E(s, m_1, \dots, m_{k-1}), m_k).$$

We say that G is *finite* if $S(G)$, $M(G)$ are finite, and all game histories have bounded length, where a game history is a sequence $h = \{m_i\}_{i=1}^{i=n}$ of moves s.t.

$$\forall 1 \leq k \leq n \ E(q_0, m_1, \dots, m_k) \text{ is defined.}$$

Note that any game history of length $2k$ can be rewritten in the form $h = \{m_i^0, m_i^1\}_{i=1}^{i=2k}$, with $m_i^j \in M_j(G)$.

It is our convention that $S(G)$ is the set of legally obtainable positions, that is positions of the form $E(q_0, m_1, \dots, m_k)$, for some m_i . For future use set $f(h) = E(q_0, m_1, \dots, m_n)$.

The above notion is more interesting when G also satisfies a certain symmetry between the two players.

Definition 9. We say that a combinatorial simultaneous game G is *symmetric* if the following conditions are satisfied.

- (1) There is a bijection $\mathcal{S} : M(G) \rightarrow M(G)$ taking $M_0(G)$ onto $M_1(G)$, and satisfying $\mathcal{S}^2 = id$, $\mathcal{S}(0) = 0$.
- (2) There is a bijection $R : B(G) \rightarrow B(G)$, satisfying $R^2 = id$ and $R(q_0) = q_0$, which we extend to $R : S(G) \rightarrow S(G)$ by $R(s, m) = (R(s), \mathcal{S}(m))$, s.t. for all $s \in S(G)$, all $m \in M(G)$, and for $\mathcal{S}$ as above we have:
 - $R \circ E(R(s), \mathcal{S}(m)) = E(s, m)$, whenever both sides are defined. Furthermore, if one side is defined then so is the other.
 - R also induces a bijection from $W_0(G)$ onto $W_1(G)$ and $D(G)$ onto $D(G)$.

Lemma 10. SGo is an ssc game, that is a symmetric simultaneous combinatorial game.

Proof. Set $G = SGo$, with the latter described in Section 2. The set $M_0(G)$ corresponds to possible moves of Black and $M_1(G)$ the moves of White, as in the game description, with the pseudo pass move 0 the standard pass move.

The state space is:

$$S(G) = B(G) \sqcup B(G) \times M(G),$$

where an element of $B(G)$ is an entanglement state as in Section 2.1: a finite nonempty set of legally obtainable positions of Go , with the initial state $\{q_0\}$, for q_0 the empty position. The displayed board is its projection.

The evolution map E is then naturally defined. For example if $s \in B(G)$ and $m_0 \in M_0(G)$, $m_1 \in M_1(G)$ are moves to empty intersections of the displayed board, then $E((s, m_0), m_1) \in B(G)$ is defined to be the entanglement state produced by the resolution rules of Section 2: each branch evolves under the interleaved moves, and the resolution prunings are applied.

To see that G is symmetric, let $R : B(G) \rightarrow B(G)$ be the map that interchanges the black and white colors of the stones in every branch (on the displayed board this interchanges black and white and leaves red stones invariant). And let $\mathcal{S} : M(G) \rightarrow M(G)$ interchange black and white moves, i.e. we have a natural isomorphism $M_0(G) \rightarrow M_1(G)$, which then induces $\mathcal{S}$. Then clearly the conditions of Definition 9 will be satisfied. $\square$

4.1. Solvability. A theorem of Zermelo [18] says that for every finite sequential game either Black or White has a winning strategy or there is a draw strategy. Clearly SGo cannot have a winning strategy, furthermore unless we put an upper bound on the length of a game, or on the number of position repetitions an SGo

game length can be infinite.² So we will assume there is such a bound, which will make *SGo* a finite game.

Question 11. *Does SGo have a strategy to draw for any board size?*

Note that the answer is yes on a 2×2 board as is easy to see. Of course we have combinatorial simultaneous games with no strategy to draw, the classical example is rock-paper-scissors. However, it does have a strategy to draw on average, which is a trivial instance of the existence of a Von Neumann–Nash equilibrium, for “matrix payoff games” [23]. *SGo* is not presented as a matrix payoff game, but passing to pure strategies, as in the proof of the theorem below, exhibits its normal form as a matrix payoff game, for a very monstrous matrix. In general:

Theorem 12. *Assigning any payoff for a win and draw, a finite simultaneous game G in the sense above has a Nash equilibrium.*

Proof. For a non final $s \in B(G)$ set

$$M_i(s) = \{m \in M_i(G) \mid E(s, m) \text{ is defined}\},$$

which is non-empty as it contains the pseudo pass move 0.

A **pure strategy** of player P_i assigns to each non final $s \in B(G)$ a move in $M_i(s)$, so that the set of pure strategies is the finite set

$$\Sigma_i = \prod_s M_i(s),$$

the product over the non final elements of $B(G)$. The space of mixed strategies of player P_i is then:

$$\mathcal{P}_i = \Delta(\Sigma_i),$$

the space of probability distributions over Σ_i , which is a compact convex simplex in $\mathbb{R}^N$, for some very large N .

A pair of pure strategies determines a play of the game, and so a payoff, by finiteness of G .³ The expected payoff function on $\mathcal{P}_0 \times \mathcal{P}_1$ is then bilinear, in particular continuous. Moreover, for a fixed strategy of the opponent, the expected payoff of P_i is a linear function on the simplex $\mathcal{P}_i$, so that the set of countering strategies of P_i is a non-empty, closed and convex face of $\mathcal{P}_i$. Consequently, the multi valued function ϕ , assigning to a joint strategy its set of countering joint strategies, has a closed graph, and the image by ϕ of any point is non-empty and convex.

Then as argued by Nash [11] using Kakutani’s fixed point theorem [10] we get existence of an equilibrium point. $\square$

Corollary 13. *A finite ssc game G has a strategy to draw on average, and in particular SGo has such a strategy.*

Proof. Assign 1 for a win, -1 for a loss and 0 to a draw. By the proof of Theorem 12, we may view G as a zero sum matrix payoff game, with the rows and columns of the payoff matrix indexed by the pure strategies Σ_0, Σ_1 . By the classical theory of matrix payoff games (see for instance [12]), all Nash equilibria have the same expected payoff for Black, namely the value v of the matrix game, and an equilibrium strategy of Black guarantees them expected payoff at least v against every strategy

²This is also true for *Go* and Chess.

³Possibly finiteness condition can be relaxed.

of White. So it is enough to show that $v = 0$; that a symmetric zero sum game has value zero is classical, cf. [7], and for completeness we give the argument.

Suppose otherwise, and suppose without loss of generality that $v > 0$, so that for an equilibrium joint strategy

$$(\mathcal{D}_0, \mathcal{D}_1) \in \mathcal{P}_0 \times \mathcal{P}_1$$

Black has expected payoff $v > 0$.

Let R, S be the symmetry operators as in the proof of Lemma 10. Naturally applying these operators to the joint strategy $(\mathcal{D}_0, \mathcal{D}_1)$ we get an equilibrium joint strategy $(\mathcal{D}'_0, \mathcal{D}'_1)$ where White has expected payoff v , that is with Black expected payoff $-v < 0$. As the latter must again be v , we have a contradiction. $\square$

4.2. Symmetric combinatorial sequential games. Combinatorial sequential games have a beautiful Conway recursive definition [4]. However, for our purpose, particular for the universal simultaneization construction, it is convenient to interpret them in terms of finite automata.

Definition 14. A *combinatorial sequential game* (with a pseudo pass move) G with 2 players P_0, P_1 consists of:

- A deterministic automaton G , whose set of states is a set $S(G)$ called the set of game states of G .
- The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \cup M_1(G)$: interpreted as possible moves of the players.
- $M_0(G) \cap M_1(G)$ has one move denoted as 0 and called the *pseudo pass move* (it has multi grading both 0 and 1).
- A $\mathbb{Z}_2$ graded set of game states $S(G) = \text{States}(G) = S_0(G) \sqcup S_1(G)$, we let $\deg : S(G) \rightarrow \mathbb{Z}_2$ denote the associated function.
- Nonempty subset $F(G) = W_0(G) \sqcup W_1(G) \sqcup D(G) \subset S(G)$ of final (accepting) states, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in S_0(G)$, understood as the state in which the game is initialized.
- A partial game evolution function:

$$E = E_G : S(G) \times M(G) \rightarrow S(G),$$

which has the properties:

- (1) $E(s, m)$ is undefined for s in one of the final states.
- (2) $E(s, m)$ is defined only if m has the same grading as s or if $m = 0$.
- (3) If $E(s, m)$ is defined then $\deg(E(s, m)) = \deg(s) + 1$.
- (4) If s is not a final state then $E(s, 0)$ is defined.
- (5) If $E(s, 0) = E(s', 0)$ then $s = s'$.

We recursively define:

$$E(s, m_1, \dots, m_k) = E(E(s, m_1, \dots, m_{k-1}), m_k).$$

As before it will be understood that $S(G)$ is the set of legally obtainable positions, that is positions of the form $E(s, m_1, \dots, m_k)$, for some m_i .

Example 15. Note that Go has a pseudo pass move, but chess does not, however chess can be augmented with a pseudo pass move, by also adjusting the rules so that king capture is legal and ends the game, while also removing the check rules.

This does not change the early-mid game dynamics at all, but the end game does significantly change, with more tie possibilities (mainly, if a player is in Zugzwang they may just pass). We will call this **augmented chess** denoted by $\widetilde{\text{chess}}$.

Definition 16. We say that a combinatorial sequential game G is **symmetric** if the following conditions are satisfied.

- (1) There is a bijection $\mathcal{S} : M(G) \rightarrow M(G)$ taking $M_0(G)$ onto $M_1(G)$ and satisfying $\mathcal{S}^2 = id$, $\mathcal{S}(0) = 0$.
- (2) There is a bijection $R : S(G) \rightarrow S(G)$, satisfying $R^2 = id$ and taking $S_0(G)$ to $S_1(G)$ s.t. for all $s \in S(G)$, all $m \in M(G)$ and for $\mathcal{S}$ as above we have:

$$R \circ E(R(s), \mathcal{S}(m)) = E(s, m),$$

whenever both sides are defined. Furthermore, if one side is defined then so is the other.

- (3) R also induces a bijection from $W_0(G)$ onto $W_1(G)$ and $D(G)$ onto $D(G)$.

Example 17. Of course $\widetilde{\text{chess}}$ and classical Go are examples of symmetric games. In case of Go, $\mathcal{S}$ interchanges black and white moves and R likewise interchanges the color of the stones on the board and shifts the initiative (it is easy to see that this does result in a legally reachable position).

It will be the assumption from now on that all sequential games that appear are symmetric (with $\mathcal{S}$, R implicitly fixed).

4.3. Universal simultaneization. Let G be sequential game (by our standing assumption with a pass move 0 and symmetric). Let $\{m_i\}$ be a sequence of game moves. If some $m_i = 0$ and $m_k \neq 0$ for some $k > i$, then we say that it is an **interior pass move**. We say that a finite sequence $\{m_i\}$ of moves of G is **pseudo legal** if it becomes a game history after adding or removing some interior pass moves from the sequence. It is not hard to see that given a pseudo legal sequence h , there is a unique minimal length game history as above, and we denote it by $leg(h)$.

For $a \in S(G)$ if there exists an $a' \in S(G)$ s.t. $E_G(a', 0) = a$ we set $\bar{a} = a'$, otherwise set $\bar{a} = E_G(a, 0)$. Property 5 in Definition 14 ensures that it doesn't matter which a' is chosen when it exists. For $a \in S(G)$ and $m \in M(G)$ set

$$a \# m = \begin{cases} E_G(a, m), & \text{if this is defined} \\ E_G(\bar{a}, m), & \text{if this is defined} \\ \text{undefined}, & \text{otherwise.} \end{cases}$$

Define $P(a, m_0, m_1) \in 2^{S(G)}$ by the following prescription in decreasing priority, where the expression is defined if and only if all the terms are defined.

- $P(a, m_0, m_1) := a \# m_0 \# m_1 \cup a \# m_1 \# m_0$ (note that if $m_i = 0$ then the pair of elements coincide so that $P(a, 0, 0) = a \# 0 \# 0$.)
- $P(a, m_0, m_1) := a \# m_0 \# m_1 \cup a \# m_1$.
- $P(a, m_0, m_1) := a \# m_0 \cup a \# m_1 \# m_0$.
- $P(a, m_0, m_1) := a \# m_0 \cup a \# m_1$.
- $P(a, m_0, m_1)$ is undefined.

We then get a partial function

$$P : S(G) \times M_0(G) \times M_1(G) \rightarrow 2^{S(G)}.$$

Definition 18. Let G be a sequential game. Its **universal simultaneization** is the simultaneous game $\text{Sim}(G) = G'$ satisfying the following.

- (1) $B(G') = 2^{S_0(G)}$. (The sub elements of $s \in \mathfrak{s} \in B(G')$ will be called **state branches**.)
- (2) The initial state $q'_0 = \{q_0\}$.
- (3) $M(G') = M(G)$ as $\mathbb{Z}_2$ graded sets.
- (4) Let $\mathfrak{s} \in B(G')$, and $m_i \in M_i(G')$ then $E_{G'}(\mathfrak{s}, m_i)$ is defined iff for each $s \in \mathfrak{s}$ s is not final and $s \# m_i$ is defined.
- (5) For $\mathfrak{s}$ and m_i as just above:

$$E_{G'}(\mathfrak{s}, m_0, m_1) := \cup_{s \in \mathfrak{s}} P(s, m_0, m_1),$$

(undefined if some $P(s, m_0, m_1)$ is undefined).

(6)

$$\mathfrak{s} \in F(G') \iff \exists s \in \mathfrak{s} s \in F(G).$$

Note that $\text{Sim}(G)$ is uniquely defined by the above conditions up to specifying the sets $W_1(\text{Sim}(G))$, $D(\text{Sim}(G))$; these will not play a role but we can specify them as follows:

-
- $\mathfrak{s} \in W_0(G') \iff \exists s \in \mathfrak{s} (s \text{ is winning for } P_0) \wedge \nexists s \in \mathfrak{s} (s \text{ is winning for } P_1).$
- Similarly,
- $\mathfrak{s} \in W_1(G') \iff \exists s \in \mathfrak{s} (s \text{ is winning for } P_1) \wedge \nexists s \in \mathfrak{s} (s \text{ is winning for } P_0).$
- In all other cases, $\mathfrak{s} \in D(G')$.

Remark 19. The passage from positions to sets of positions is a form of the classical subset construction of automata theory [14]. In game theoretic terms, a state of $\text{Sim}(G)$ is a knowledge state: the set of classical positions consistent with what the players have observed. Analogous knowledge based constructions are used in the study of games of imperfect information, see for instance [16], [5].

Lemma 20. $\text{Sim}(G)$ is symmetric.

Proof. Set $G' = \text{Sim}(G)$, and let $\mathcal{S}, R$ be the symmetry operators of G . Set $\tilde{\mathcal{S}} : M(G') \rightarrow M(G')$ to just be $\mathcal{S}$. For

$$\{s_0, \dots, s_k\} = \mathfrak{s} \in B(G'),$$

set

$$\tilde{R}(\mathfrak{s}) = \{R(s_0), \dots, R(s_k)\}.$$

For

$$(\mathfrak{s}, m) \in B(G') \times M(G') \subset S(G'),$$

set

$$\tilde{R}(\mathfrak{s}, m) = (\tilde{R}(\mathfrak{s}), \tilde{\mathcal{S}}(m)).$$

The needed symmetry properties for the maps $\tilde{\mathcal{S}}, \tilde{R}$ are then verified using the symmetry properties of $\mathcal{S}, R$ as follows. It is easily seen that $R(\bar{a}) = R(\bar{a})$, hence

$$R(a) \# S(m) = \begin{cases} E_G(R(a), S(m)) = R \circ E_G(a, m), & \text{if defined} \\ E_G(R(\bar{a}), S(m)) = R \circ E_G(\bar{a}, m), & \text{if defined} \end{cases}$$

and so

$$(21) \quad R(a) \# S(m) = R(a \# m),$$

furthermore if one side is defined then so is the other. Suppose that $\mathfrak{s} \in B(G')$ and $m_i \in M_i(G')$ then we get:

$$\begin{aligned} \tilde{R}(E_{G'}(\tilde{R}(\mathfrak{s}), \tilde{S}(m_0), \tilde{S}(m_1))) &= \tilde{R} \circ \cup_{s \in \mathfrak{s}} P(R(s), S(m_0), S(m_1)) \\ &= \tilde{R} \circ \cup_{s \in \mathfrak{s}} R \circ P(s, m_0, m_1) \text{ using (21)} \\ &= \cup_{s \in \mathfrak{s}} P(s, m_0, m_1) \\ &= E_{G'}(\mathfrak{s}, m_0, m_1), \end{aligned}$$

furthermore if one side is defined then so is the other. This verifies part of the symmetry property, specifically property 2 of Definition 9 is verified for $\mathfrak{s} \in B(G') \times M(G')$, the rest is similar. $\square$

Definition 22. Let G_0 be a sequential game. We say that G_1 is a *simultaneization* of G_0 if the following is satisfied.

- (1) G_1 is a symmetric simultaneous game.
- (2) $M(G_1) = M(G_0)$ as $\mathbb{Z}_2$ graded sets.
- (3) There is a map $\rho : B(G_1) \rightarrow B(\text{Sim}(G_0))$ satisfying $\rho(q_0) = q'_0$ and s.t. for $s \in B(G_1)$, $E_{G_1}(s, m_0, m_1)$ is defined iff $E_{\text{Sim}(G_0)}(\rho(s), m_0, m_1)$ is defined, and in this case

$$\rho(E_{G_1}(s, m_0, m_1)) \subset E_{\text{Sim}(G_0)}(\rho(s), m_0, m_1).$$

(This expresses the idea that evolution in G_1 is the evolution in the universal simultaneization with some branch reduction mechanism.)

- (4) $\rho(F(G_1)) \subset F(\text{Sim}(G_0))$.
- (5) We say that a state $s \in B(G_1)$ is *semi-classical* (relative to G_0) if there is finite game history $h_s = \{m_i^0, m_i^1\}$ of G_1 s.t. $f(h_s) = s$ and s.t. both $h_s = \{m_i^0, m_i^1\}$ and $h'_s = \{m_i^1, m_i^0\}$ are pseudo legal move sequences of G_0 and $f(\text{leg}(h_s)), f(\text{leg}(h'_s))$ coincide, after possibly applying the pass move $E_{G_0}(\cdot, 0)$ to one of them to match the gradings. We then ask that: for a semi-classical s $\rho(s)$ is a singleton set consisting of $f(\text{leg}(h_s))$.
- (6) If s is semi-classical and $\rho(E_{G_1}(s, m_0, m_1))$ is a singleton then $\rho(s) \# m_0 \# m_1 = \rho(s) \# m_1 \# m_0$.

For $a \in S(G_0)$ set

$$\mathbb{M}(a) = \{m \in M(G_0) \mid a \# m \text{ is defined}\},$$

which we call the *interface* of the position a : the set of moves of either player available at a , with the operation $\#$ absorbing the turn order. Likewise, for $s \in B(G_1)$ set $M(s) = \{m \in M(G_1) \mid E_{G_1}(s, m) \text{ is defined}\}$, the interface of the state s .

Lemma 23. Let G_1 be a simultaneization of G_0 . Then for every non final $s \in B(G_1)$:

$$M(s) = \bigcap_{a \in \rho(s)} \mathbb{M}(a).$$

That is, the interface of a state is the intersection of the interfaces of its branches. In particular, every branch of $\rho(s)$ is non final.

Proof. First, for $m \in M(G_1)$, $E_{G_1}(s, m)$ is defined if and only if the pair move $E_{G_1}(s, m, 0)$ (in the order of the pair prescribed by the grading of m) is defined. Indeed, the latter presupposes the former, and conversely if $E_{G_1}(s, m)$ is defined then so is $E_{G_1}(s, 0)$, as s is not final, and so by the properties of Definition 8 the pair move is defined, using that the pseudo pass move has both gradings. By condition 3, the pair move is defined if and only if it is defined in $\text{Sim}(G_0)$ at $\rho(s)$, which by the construction of $\text{Sim}(G_0)$ holds if and only if every $a \in \rho(s)$ is non final and $a \# m$, $a \# 0$ are defined. As $a \# 0$ is defined for non final a , this says exactly that $m \in \mathbb{M}(a)$ for every $a \in \rho(s)$. Applying this with $m = 0$, which lies in $M(s)$ since s is not final, shows that every branch is non final. Finally, $\rho(s)$ is non-empty, as the empty set is not a legally obtainable state of $\text{Sim}(G_0)$. $\square$

Definition 24. Let G_0 be a sequential game. We will say that a simultaneization G_1 of G_0 is a *simple simultaneization of G_0* if for every non final $s \in B(G_1)$ there is a legally obtainable position $a \in S(G_0)$ with

$$\mathbb{M}(a) = M(s),$$

where for $s = q'_0$ the initial state of G_1 , a can be taken to be q_0 .

Remark 25. Equivalently, there is a map $\phi : B(G_1) \rightarrow S(G_0)$ with $\phi(q'_0) = q_0$, s.t. for all $s \in B(G_1)$, $m \in M(G_0)$: $E_{G_1}(s, m)$ is defined if and only if $\phi(s) \# m$ is defined and s is not final. Note that no compatibility of ϕ with the game evolutions is asked for. By Lemma 23, a state s is semantically the set of branches $\rho(s)$; simplicity says that, nevertheless, the moves available at s are always the moves of a single classical position of G_0 , so that the game can be refereed classically. The dynamical content of being a simultaneization is carried by ρ , at the level of branches, and this is a natural division: requiring ϕ to also intertwine the evolutions would be too strong, as no single classical position can track the captures of SGo , so that even SGo would fail to be simple. There may certainly be other reasonable notions of “simplicity”.

Let Go be the classical game with Chinese style area scoring and standard pass move, with the following stipulations: suicide is a legal move, as for example in the Tromp–Taylor rules [22, 21], there is no Ko or superko rule, and the game is ended analogously to the way SGo is ended, so that the role of a superko rule is played by the end of the game rules.

Theorem 26. SGo is a simple simultaneization of Go .

Proof. The first two properties are by construction and Lemma 10.

Let $\rho : B(SGo) \rightarrow B(\text{Sim}(Go))$ be the tautological map: an entanglement state is a finite set of positions of Go , that is an element of $B(\text{Sim}(Go))$ (we suppress the routine matching of the pass gradings). Then $\rho(q_0) = \{q_0\} = q'_0$. The branch evolution of Section 2.1 is precisely the evolution by the map P above: playing the two moves in both orders, with a move to an occupied intersection forfeited, is the prescription defining P . So $E_{SGo}(s, m_0, m_1)$ is obtained by applying the resolution prunings to $E_{\text{Sim}(Go)}(\rho(s), m_0, m_1)$, and the inclusion in condition 3 holds by construction. For the definedness: a move of SGo is available if and only if its intersection is empty on the displayed board, if and only if, by the definition of the projection, that intersection is empty in every branch, if and only if the move is defined in $\text{Sim}(Go)$ at $\rho(s)$. Condition 4 holds by the end of game rules.

For the semi-classicality condition: along a game history as in the definition of a semi-classical state no two moves interfere, so each branch evolution produces a single position, and the entanglement state remains a singleton, which the prunings preserve, being nonempty subsets by Lemma 6; so $\rho(s)$ is the required singleton. Condition 6 holds since whenever both composites are defined, the moves did not interfere and the composites agree.

To verify the simplicity property: since in our version of *Go* suicide is a legal move and there is no Ko rule, for any non final position a of *Go* the interface $\mathbb{M}(a)$ consists of the pass moves and the moves of either player to the empty intersections of a . Likewise, for a non final $s \in B(SGo)$, $M(s)$ consists of the pass moves and the moves of either player to the empty intersections of s . So it suffices to exhibit a legally obtainable position $\phi(s)$ of *Go* with the same empty intersections as s . Let $\phi(s)$ be obtained by replacing every stone of s by a black stone. To see that $\phi(s)$ is legally obtainable, fix an empty intersection e of s , which exists as s is not final, and let Black play the stones of $\phi(s)$ in order of weakly decreasing distance to e , while White always passes. When a stone is placed, a geodesic from it to e exits the set of already placed stones at an empty intersection adjacent to the set, so at every moment every black component has a liberty, and no captures or suicides occur. $\square$

It is natural to ask how much of the entanglement state the board itself determines:

Question 27. *To what extent is the entanglement state of a legal SGo history determined by its sequence of displayed boards? In all our computer experiments, two legal histories displaying the same boards had nested entanglement states, so that the smaller state could serve for both.*

Question 28. *Does augmented chess as in Example 15 have a simple simultaneization?*

We will show that the answer is no, whatever the branch reduction mechanism. The following observation isolates the role of the symmetry of G_1 .

Lemma 29. *Let G_1 be a simultaneization of G_0 , and suppose that $s \in B(G_1)$ is obtained from the initial state by a game history consisting of mirror pairs, that is, of simultaneous pairs of the form $(x_i, \mathcal{S}(x_i))$. Then $R(s) = s$, and consequently $\mathcal{S}(M(s)) = M(s)$.*

Proof. The mirror of the pair $(x, \mathcal{S}(x))$ is the pair $(\mathcal{S}(\mathcal{S}(x)), \mathcal{S}(x)) = (x, \mathcal{S}(x))$ itself. So applying the symmetry property of Definition 9 along the game history, together with $R(q_0) = q_0$, we get $R(s) = s$. Then for any m , $E_{G_1}(s, m)$ is defined if and only if $E_{G_1}(R(s), \mathcal{S}(m)) = E_{G_1}(s, \mathcal{S}(m))$ is defined, giving $\mathcal{S}(M(s)) = M(s)$. $\square$

Theorem 30. *$\widetilde{\text{chess}}$ has no simple simultaneization. In particular, the universal simultaneization $\text{Sim}(\widetilde{\text{chess}})$ is not simple.*

Proof. Let G_1 be any simultaneization of $\widetilde{\text{chess}}$, with P_0 the White player, and consider the game history of G_1 consisting of the mirror pairs

$$(a4, a5); (Ra3, Ra6); (Rb3, Rb6),$$

8	.	n	b	q	k	b	n	r
7	.	p	p	p	p	p	p	p
6	.	r	.	.	.	.	.	.
5	p	.	.	.	.	.	.	.
4	P	.	.	.	.	.	.	.
3	.	R	.	.	.	.	.	.
2	.	P	P	P	P	P	P	P
1	.	N	B	Q	K	B	N	R
	a	b	c	d	e	f	g	h

FIGURE 1. The position a^* , reached by the mirror pairs $(a4, a5)$; $(Ra3, Ra6)$; $(Rb3, Rb6)$. White pieces are in capitals. The conflicting simultaneous moves are the rook moves $m_0 : b3 \rightarrow b5$ and $m_1 = \mathcal{S}(m_0) : b6 \rightarrow b4$.

in standard (except for the pass move usage) chess algebraic notation. The moves of each pair do not interact, so both interleavings are pseudo legal with the same outcome, and each state along this history is semi-classical. By the semi-classicality condition of Definition 22, applied inductively along the history, the history is legal in G_1 and for the resulting state $s^* \in B(G_1)$ we have $\rho(s^*) = \{a^*\}$, with a^* the classical position of Figure 1. Note that this pins down $\rho(s^*)$ for every simultaneization G_1 , whatever its branch reduction mechanism.

Now let m_0 be the White rook move $b3 \rightarrow b5$, and $m_1 = \mathcal{S}(m_0)$ the Black rook move $b6 \rightarrow b4$. Each order of these moves blocks the other: after $b3 \rightarrow b5$ the Black rook's path is blocked at $b5$, while after $b6 \rightarrow b4$ the White rook's path is blocked at $b4$. So in $\text{Sim}(\widetilde{\text{chess}})$ this pair resolves by the lowest priority case in the definition of P , and

$$P(a^*, m_0, m_1) = \{a_0, a_1\},$$

where in a_0 the White rook stands on $b5$ with the Black rook on $b6$, and a_1 is the mirror image of a_0 . By condition 3, $s_1^* = E_{G_1}(s^*, m_0, m_1)$ is defined and $\rho(s_1^*)$ is a non-empty subset of $\{a_0, a_1\}$. Moreover s_1^* is not final, by condition 4, as neither a_0 nor a_1 is final in $\widetilde{\text{chess}}$.

Suppose that $\rho(s_1^*)$ is a singleton, say $\{a_0\}$. The history of s_1^* consists of mirror pairs, so by Lemma 29 $M(s_1^*)$ is $\mathcal{S}$ invariant. But by Lemma 23, $M(s_1^*) = \mathbb{M}(a_0)$, which is not $\mathcal{S}$ invariant: in a_0 White has rook moves from $b5$, while Black has no rook moves from $b4$. The case of $\{a_1\}$ is mirrored. Thus this branch reduction is forbidden by the symmetry of G_1 , and $\rho(s_1^*) = \{a_0, a_1\}$.

By Lemma 23, $M(s_1^*) = \mathbb{M}(a_0) \cap \mathbb{M}(a_1)$, which is computed directly: for each player it consists of the pass move, the moves of the c through h pawns, and the four knight moves. There are no queen side rook moves, as these rooks stand on different squares in a_0 and a_1 , and no a or b pawn moves. Now suppose that G_1 is simple, so that there is a legally obtainable position c of $\widetilde{\text{chess}}$ with $\mathbb{M}(c)$ equal to this set. The interface $\mathbb{M}(c)$ forces every piece of c other than the two queen side rooks, the two a pawns and the two b pawns to stand on its starting square. One now checks the finitely many placements, or absences by capture, of the six remaining pieces, and each fails. For example, if the White b pawn is at $b2$, then $b3$ must be occupied, necessarily by a queen side rook or the Black b pawn, giving

respectively a legal rook move from $b3$ and the legal capture $b3 \times c2$, neither of which is in the set; while if the White b pawn is absent, having been captured, and $b2$ is unoccupied, then the $c1$ bishop has the legal move to $b2$, again not in the set. So we have a contradiction, and G_1 is not simple. $\square$

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